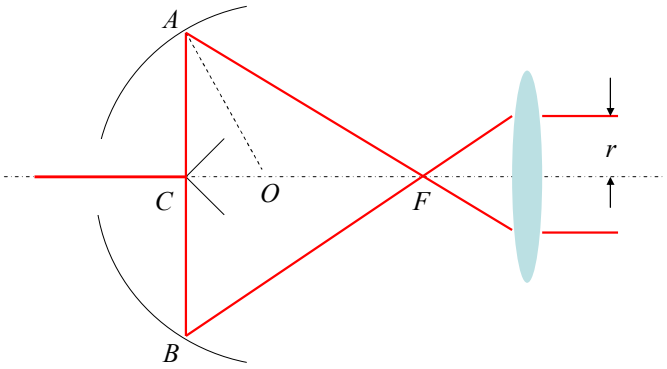
Solution:

1、 Below, rays reflected by the spherical mirror meet at the convex lens focus.



2、 The ray diagram gives $CO = x$. Let $\angle CAO = \alpha$; geometry gives:

$$\sin\alpha = \frac{x}{R}, \tag{1}$$

$$r = f\cot(2\alpha) \quad . \tag{2}$$

Combining the two equations gives

$$x = R\sin(\frac{1}{2}\arctan\frac{f}{r}). \tag{3}$$

Grading: Correct ray diagram: 5 points; equation （3）: 5 points.